EXTENSION UNIT: SHOP DRAWINGS

Three-drawer base unit under the bench extension. Plywood carcass, plinth and drawers on Hafele Alto runner drawbars.

PARTS SCHEDULE (15 mm plywood; grain along the length)

Ref	Part	Qty	Length	Width	Thick	Notes
C1	Side, free end	1	749	522	15	Full height
C2	Side, joint end	1	749	522	15	Two 6.5 wide link-bolt slots, open to the top edge
C3	Bottom	1	775	507	15	Sits on the plinth between the sides
C4	Back	1	775	749	15	Full height, between the sides on the ends of the plinth side rails; C3 butts its front face
C9	Top rail, front	1	775	60	15	Between the sides at the top, under the benchtop
P1	Plinth rail, front	1	767	125	15	Pocket-screwed up into C3
P2	Plinth rail, rear	1	767	125	15	Fully under C3; pocket-screwed up into it
P3, P4	Plinth rail, side	2	461	125	15	P3 free end, P4 joint end; screwed up into the sides
Kb	Kickboard cover	1	852	115	15	Screwed on from inside the plinth
D1, D2	Drawer front	2	802	212.7	15	45 deg mitre top (R3 nose, R1 sanded), 15 high bottom mitre; no base groove
D3	Drawer front	1	802	326.3	15	45 deg mitre top (R3 nose); bottom level with the existing doors
D1, D2	Drawer side	4	495	153	15	Through base groove; pocket holes; fixing details
D3	Drawer side	2	495	231	15	As D1, D2
D1, D2	Drawer back	2	713	148	15	Base groove 15.5 x 8; 5 above side bottoms
D3	Drawer back	1	713	226	15	As D1, D2
TS	Top strip	1	802	32.3	15	False front above D1: bottom mitre as D1, D2; plain top
TP	Strip packer	1	775	15	6	6 mm ply or hardwood; glued behind TS and to C9
D1-D3	Drawer base	3	728.5	487.5	15	Butts the front; rear edge in the back groove

HARDWARE AND FIXINGS

Hafele Alto 86 x 500 mm set	3	Runners, drawbars, front brackets and snap clips; steel sides not used
Snap clips (in the Alto set)	6	2 per drawer; fixing option 1
No.6 x 12 pan-head screws	12	Option 2 drawbar fixing (6 for option 1 clip screws)
Pocket-hole screws, 25 mm, coarse	70	Drawer boxes 28; carcass and plinth 42 (C3 8, C4 10, C9 4, plinth rail joints 8, plinth to carcass 12)
Steel angle brackets 25 x 25 x 2, 40 long	9	Benchtop: 3 each side, 3 on the back
M6 x 65 bolts, washers	2	Link to the main run's framing rail
No.8 x 25 screws	4	Kickboard cover
No.8 x 12 and No.8 x 16 screws	18 + 18	Benchtop brackets: x12 into the carcass, x16 up into the benchtop

GENERAL NOTES

1. Material: 15 mm structural CD plywood (Bunnings STRUCTAply); all parts 15 mm. Cut square to +/-0.5 mm.
2. Hand routing only. No CNC. Base slot 15.5 wide x 8 deep, lower edge 13 above the bottom edge of the side: through the sides and across the back.

Build each box front-last: sides and back first, slide the base in from the front, then fit the front.

3. Drawer pulls: a 45 degree mitre on the front 7.5 mm of the top edge of each front (R3 nose) with a 7.5 mm catch wall behind it (half-thickness rebate from the back), and a 45 degree mitre (15 mm high on the face, about 15 mm clear to the mitre below) on the bottom edge of D1 and D2. The nose of each front stays 5 mm below the lowest point of the front above (after rounding), so the fronts overlap by 10 mm and each drawer opens without catching the one below.

Pull profiles and the gaps between fronts are on sheet 9.

4. Drawer fixing shown: option 1 (snap clip + rear tab notch). Option 2 (screws from below), sheet 8.
5. Runners: the steel stack hangs 45 mm below each drawbar plate, drawbar outer face 14 mm from the cabinet wall,

8 mm underside opening facing away from the wall. Box sides sit on the 15.2 mm drawbar flat, 16 mm from the wall.

6. Drawer fronts: bracket holes to the Hafele front pattern (datasheet page 1); not dimensioned here.
7. The fixed runner is modelled as a 14 x 45 block; the real section is not drawn. Check it on the first fitted drawer.
8. Benchtop (existing placeholder) rests on the carcass sides, back and front top rail C9; 9 angle brackets fix it (sheet 3).
9. Carcass and plinth joints: glued pocket-screw butt joints, 25 mm coarse screws (42 in all, positions in the guide); 12 of them fix the plinth P1 to P4 up into C3 and the sides from inside the frame.
10. Unit height floor to benchtop underside 874 (top surface 900 with the 26 mm timber top). The extension top has a 12 mm hem under the front edge only and overhangs the carcass 38 mm at the front. Plinth frame 125 (kickboard cover 115, as the existing kickboard). Drawer fronts overlay the carcass 1.5 mm in from each end.
11. Carcass and plinth stop 40 mm short of the wall to clear a skirting board (20 to 25 projection). The benchtop keeps its full 600 depth.

SHEET INDEX

1 Cover, schedules
2 General arrangement: front elevation, side section A-A
3 Plan and cross-section B-B
4 Carcass parts C1 to C4 and C9
5 Plinth P1 to P4 and kickboard Kb
6 Drawer sides
7 Drawer fronts, backs and bases
8 Drawer box assembly and drawbar fixing options
9 Mitre pull profiles and gaps
10 Cutting diagrams (exact sizes, 2 sheets)
11 Cutting diagrams for store cuts on 100 mm multiples (3 sheets)

KITCHEN BENCH EXTENSION - EXTENSION UNIT

COVER SHEET AND SCHEDULES

Parts schedule, hardware, general notes

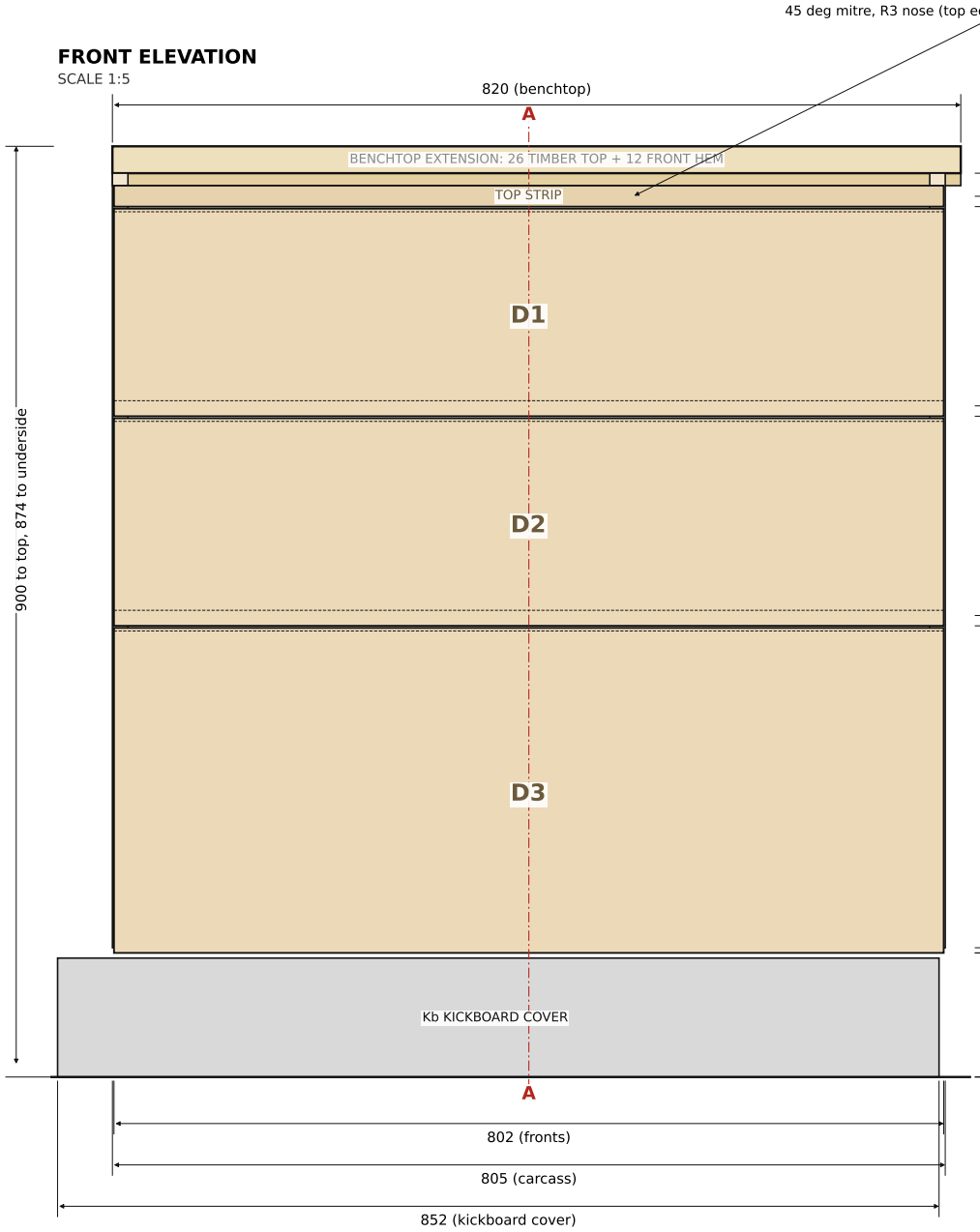
15 mm ply unless noted. Do not scale: dimensions govern.

SHEET 1

DATE 2026-11-07

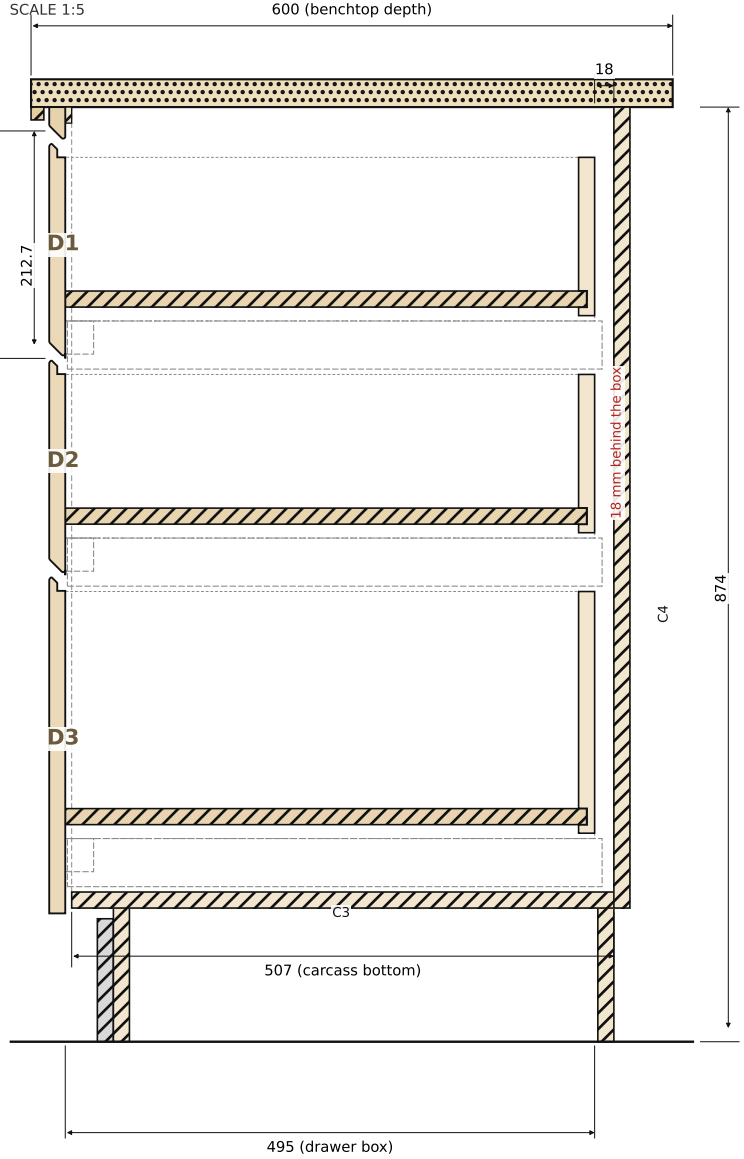
FRONT ELEVATION

SCALE 1:5



SECTION A-A (looking toward the main run)

SCALE 1:5



Drawbar plate tops (= box side bottoms): D1 h=674, D2 h=471, D3 h=190 above the floor (534 / 331 / 50 above the top of the carcass bottom).
Hidden grey: drawbar, bracket and runner block (hang 45 below each plate). Base slides in from the front into the side grooves and the back groove.

KITCHEN BENCH EXTENSION - EXTENSION UNIT

GENERAL ARRANGEMENT

Front elevation and side section A-A. Scale 1:5

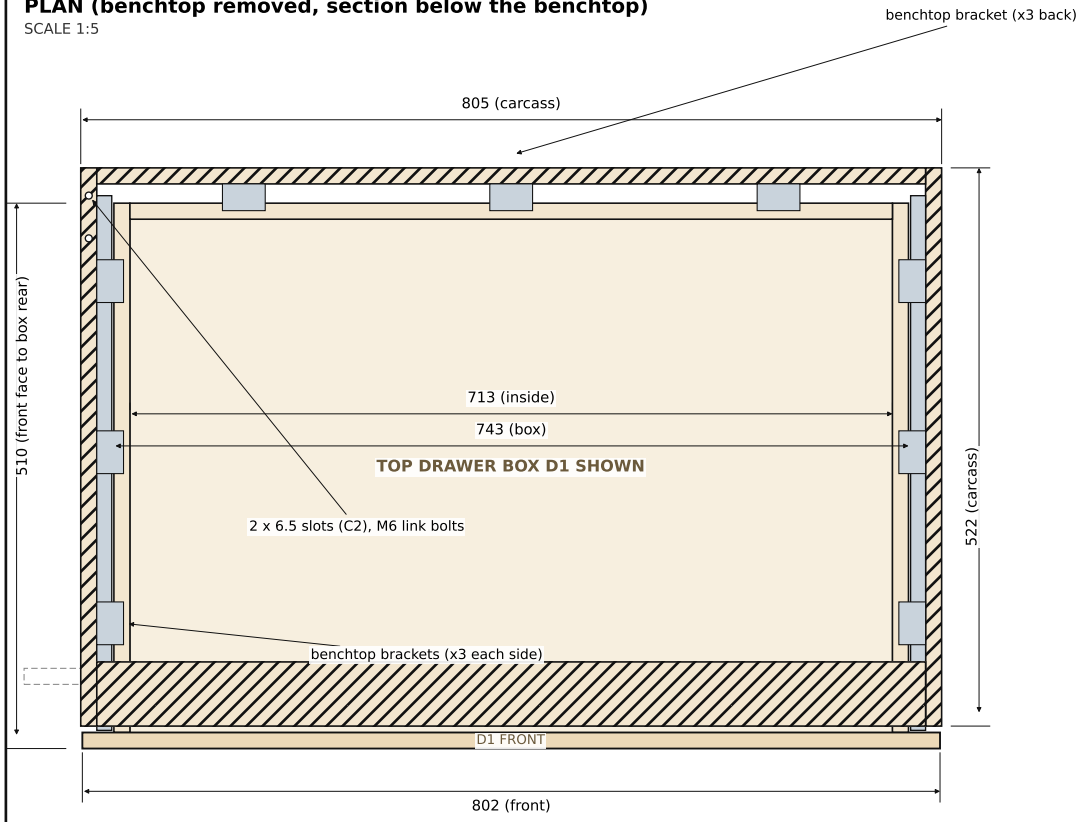
15 mm ply unless noted. Do not scale: dimensions govern.

SHEET **2**

DATE 2026-11-07

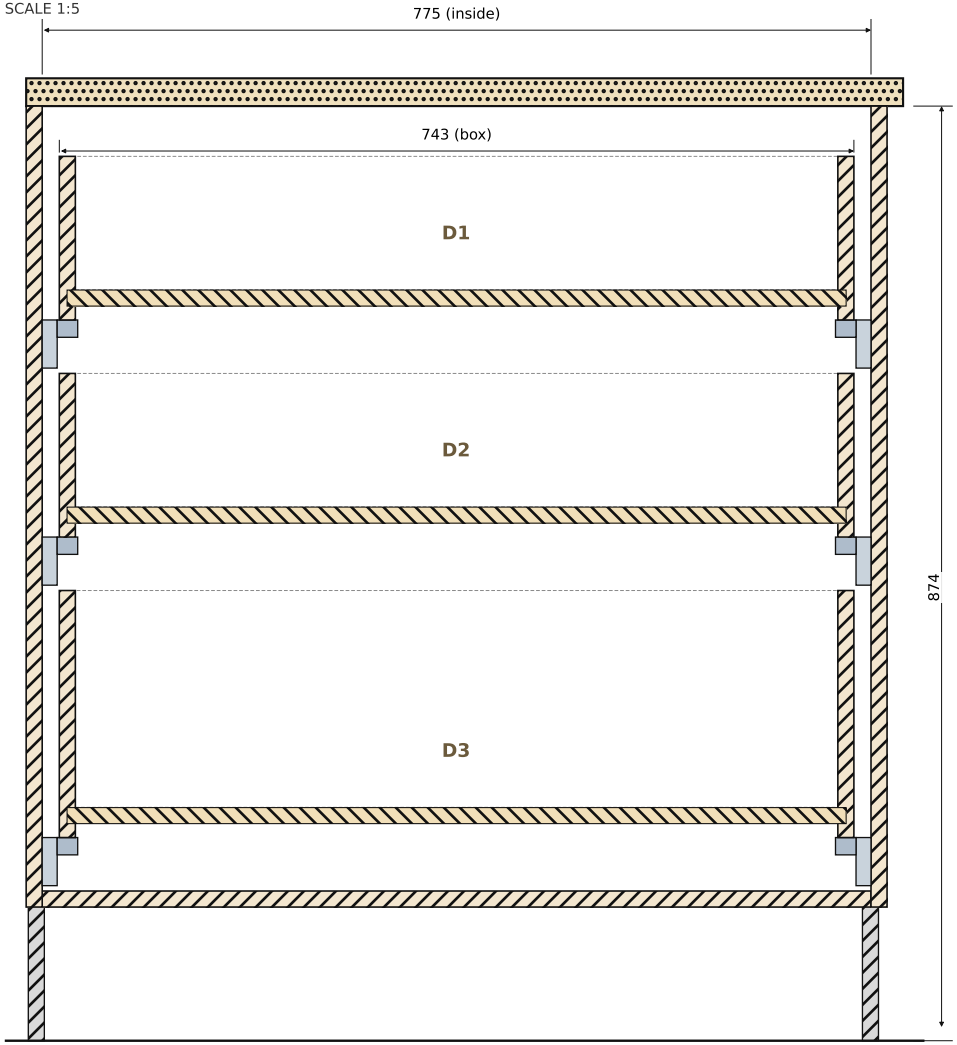
PLAN (benchtop removed, section below the benchtop)

SCALE 1:5



SECTION B-B at 300 from the benchtop edge

SCALE 1:5



Grey blocks: runner (14 x 45) and drawbar. Boxes sit on the drawbar flats, outer faces 16 from the walls.

KITCHEN BENCH EXTENSION - EXTENSION UNIT

PLAN AND CROSS-SECTION B-B

Plan with benchtop removed, and section across the drawers. Scale 1:5

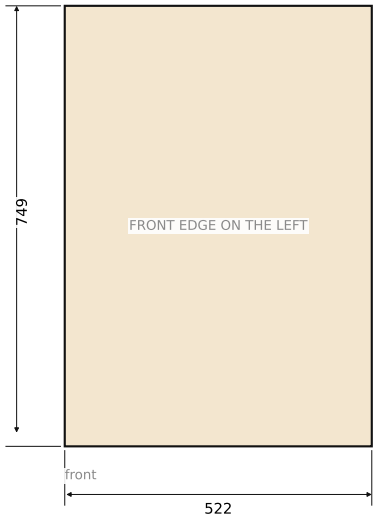
15 mm ply unless noted. Do not scale: dimensions govern.

SHEET **3**

DATE 2026-11-07

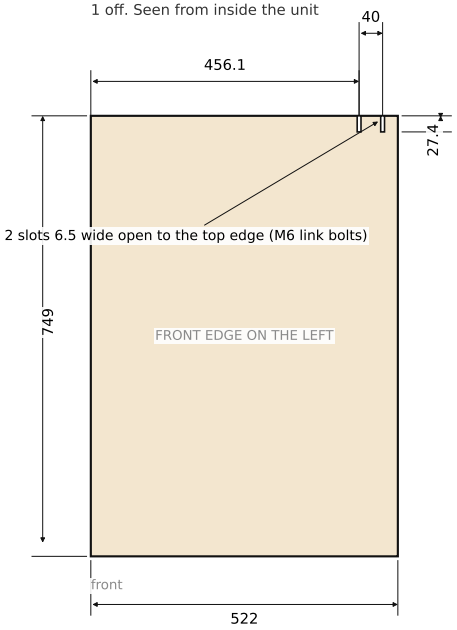
C1 SIDE, FREE END

1 off. Seen from outside



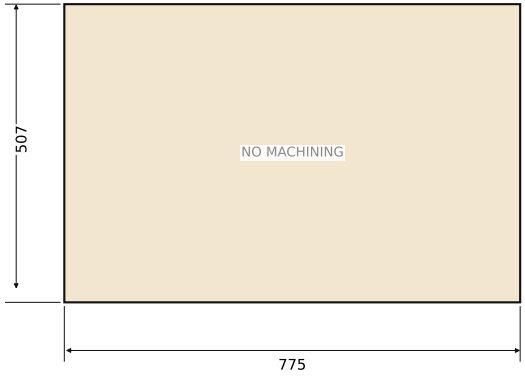
C2 SIDE, JOINT END

1 off. Seen from inside the unit



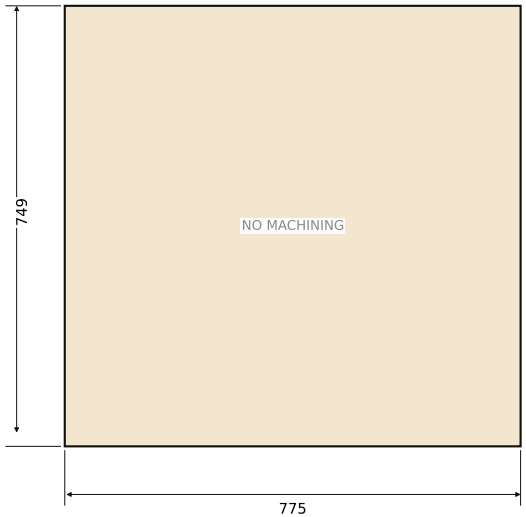
C3 BOTTOM

1 off. Plan, front edge at the bottom



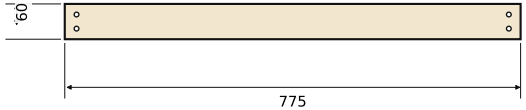
C4 BACK

1 off. Seen from inside



C9 TOP RAIL, FRONT

1 off. Between the sides at the top; 2 pocket holes at each end



ASSEMBLY

Sides C1, C2 stand on the plinth. C3 goes between them, its bottom edge flush with the bottom edges of the sides and its front edge flush with the side fronts.

C4 is full height, between the sides, standing on the ends of the plinth side rails, its rear face flush with the side rear edges; C3 butts its front face. C9 goes between the sides at the top front.

Glue every joint and pocket-screw (25 mm) from C3 (4 per side, in its underside), C4 (5 per side, in its rear face) and C9 (2 per end, in its underside) into the faces of the sides. Check square (within 1 mm).

Overall carcass: 805 wide x 522 deep x 749 high, 775 clear between the sides, inside depth 507, inside height above C3 734. The top of the carcass is 874 above the floor.

The benchtop (existing) sits on the sides and back; 9 angle brackets 25 x 25 x 2 x 40 long fix it (see sheet 3). Two M6 x 65 bolts through C2 link the unit to the main run.

KITCHEN BENCH EXTENSION - EXTENSION UNIT

CARCASS PARTS C1 TO C4

Scale 1:8. Dimensions in mm. All 15 mm ply

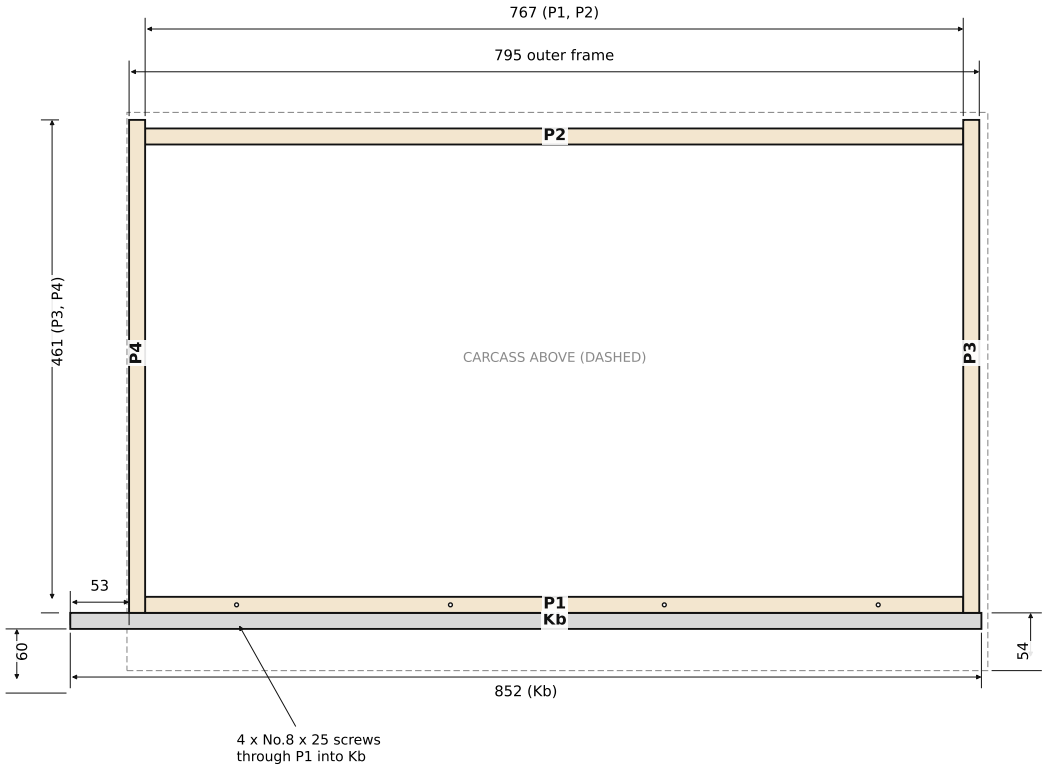
15 mm ply unless noted. Do not scale: dimensions govern.

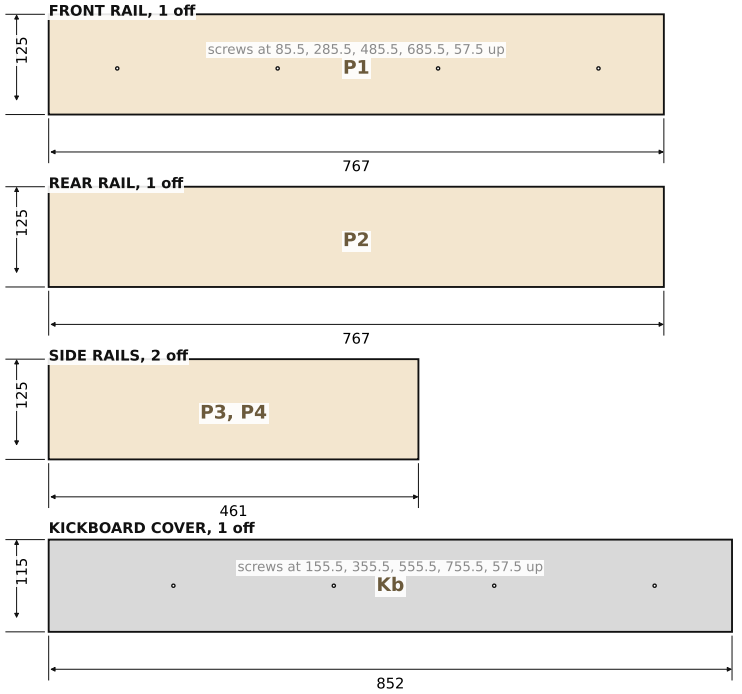
SHEET **4**

DATE 2026-11-07

PLINTH FRAME P1 TO P4 AND KICKBOARD Kb, PLAN

SCALE 1:5





Plinth: glue P1 and P2 between the two side rails P3 and P4 and pocket-screw them (25 mm, 2 per end at 30 and 85 above the floor, in the inner faces of P1 and P2); check square.

It stands on the floor and carries the carcass. Kb is a cover only: screw on from inside through P1.

Kb runs 53 past the joint end to meet the main run's kickboard; its face is 60 behind the drawer fronts.

P2 sits fully under C3 (d 530 to 545), its rear face level with C3's rear edge; P3 and P4 run 8 mm on under C4 (d to 553).

Carcass to plinth: with the carcass upside down, set the frame on it and pocket-screw from inside the frame, up into C3 (P1: 4 at 100 and 300 either side of the centre line, P2: 4 likewise) and into the bottom edges of the sides (P3, P4: 2 each, 159 and 359 back from the front edge of the benchtop).

KITCHEN BENCH EXTENSION - EXTENSION UNIT

PLINTH P1 TO P4 AND KICKBOARD Kb

Scale 1:5. Dimensions in mm. All 15 mm ply

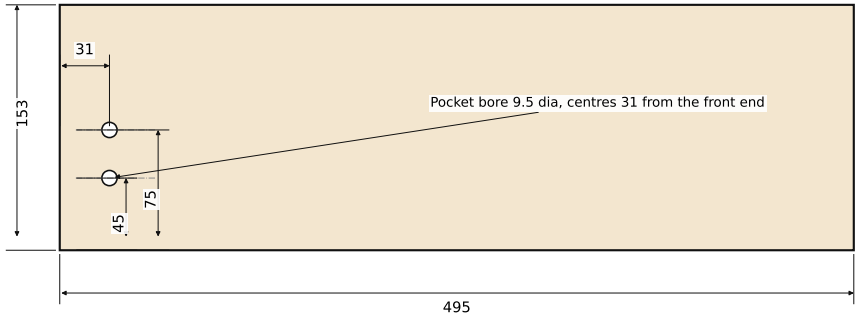
15 mm ply unless noted. Do not scale: dimensions govern.

SHEET 5

DATE 2026-11-07

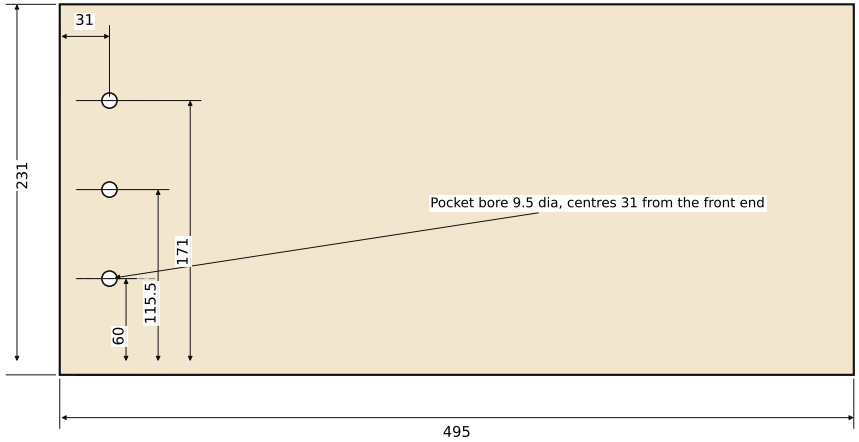
D1, D2 SIDE (153 HIGH), 4 OFF: OUTER FACE

pocket holes for the front (15 deg, 25 mm screws)



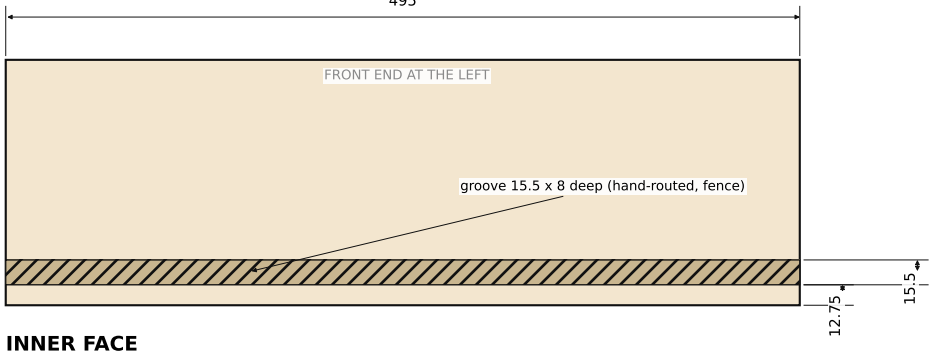
D3 SIDE (231 high), 2 OFF: OUTER FACE

pocket holes for the front (15 deg, 25 mm screws)



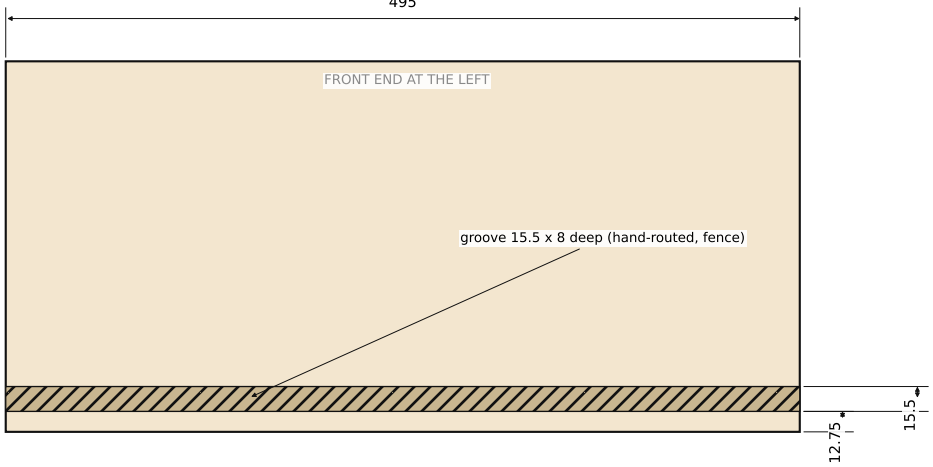
INNER FACE

base groove, through, open at both ends



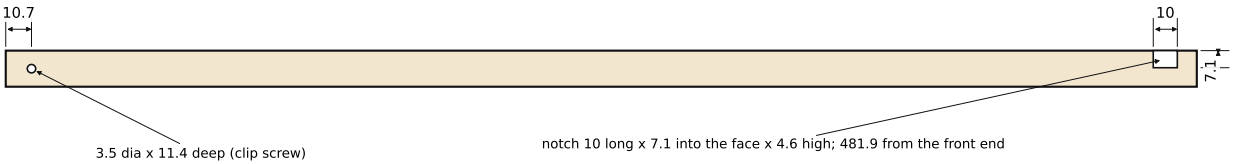
INNER FACE

base groove, through, open at both ends



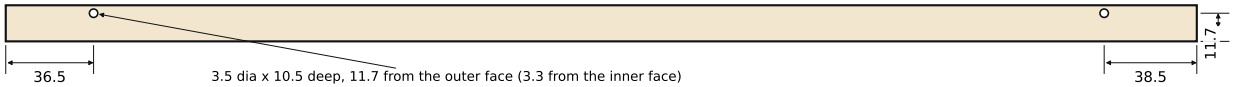
OPTION 1 (snap clip): BOTTOM EDGE

left side shown; the right side is a mirror (notch on the inner face side)



OPTION 2 (screws from below): BOTTOM EDGE

mark through the holes drilled in the drawbar



KITCHEN BENCH EXTENSION - EXTENSION UNIT

DRAWER SIDES

Scale 1:3.3. 6 off: 4 x D1/D2 (153 high), 2 x D3 (231 high). 15 mm ply. Pairs are mirrored

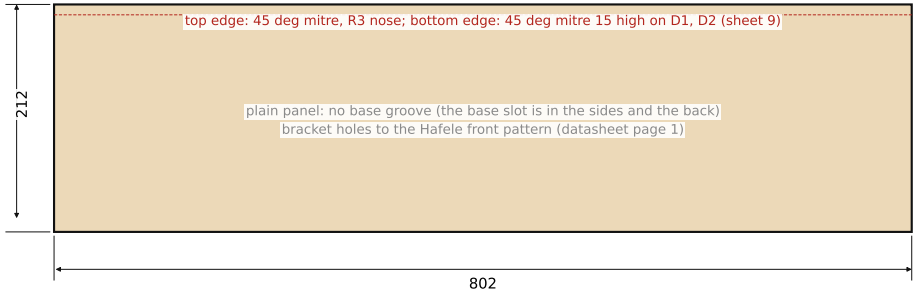
15 mm ply unless noted. Do not scale: dimensions govern.

SHEET 6

DATE 2026-11-07

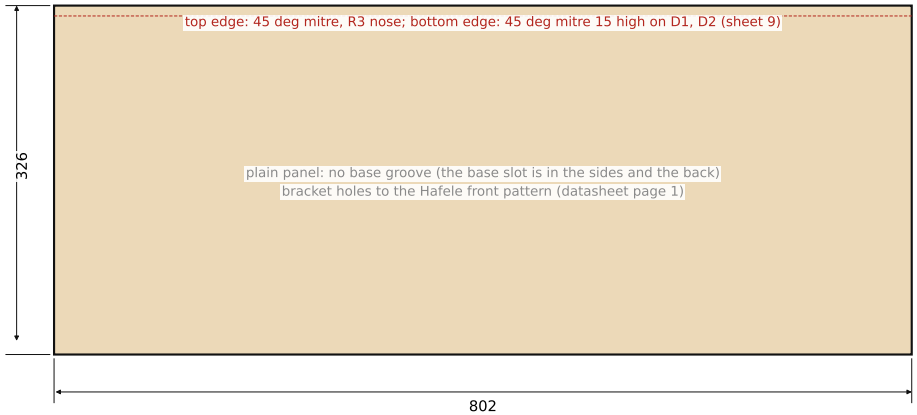
D1, D2 FRONT (2 off)

back face. Top edge toward the top of the sheet



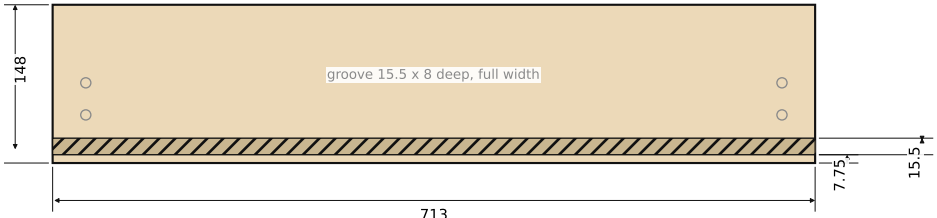
D3 FRONT (1 off)

back face. Top edge toward the top of the sheet



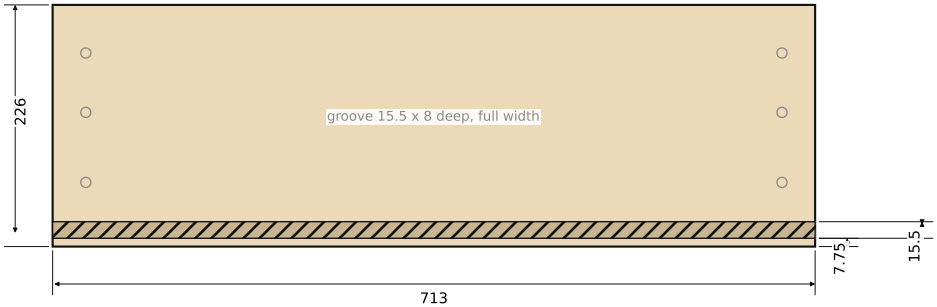
D1, D2 BACK (2 off): 713 x 148

front face (toward the base): base groove, full width, open at both ends



D3 BACK (1 off): 713 x 226

front face (toward the base): base groove, full width, open at both ends



Hidden grey circles: pocket bores in the rear face, 9.5 dia at 15 deg, 31 from each end, 25 mm screws.

Pocket heights above the bottom edge: D1, D2 at 45 and 75; D3 at 60, 125.5 and 181.

DRAWER BASE (3 off)

15 mm ply, no machining. Long side is the depth. Scale 1:6.7



Front edge butts the front panel, flush with the side ends. Rear edge ends 0.5 short of the bottom of the back groove (7.5 into the back).

KITCHEN BENCH EXTENSION - EXTENSION UNIT

DRAWER FRONTS, BACKS AND BASES

Scale 1:5. 15 mm ply. Fronts seen from the BACK face; backs seen from the FRONT face (the face toward the base)

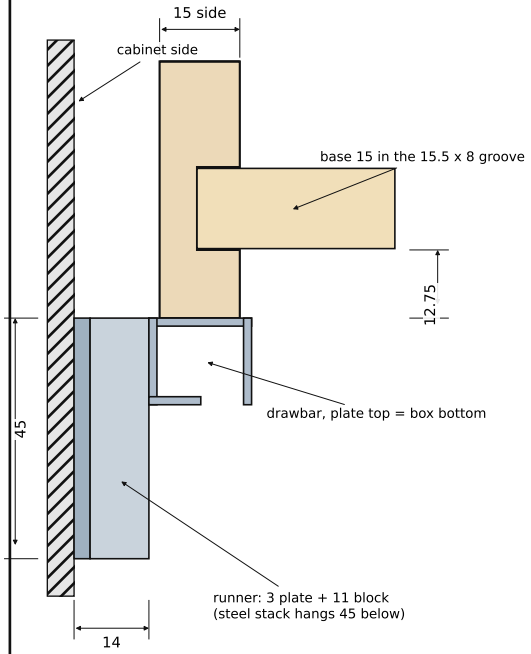
15 mm ply unless noted. Do not scale: dimensions govern.

SHEET 7

DATE 2026-11-07

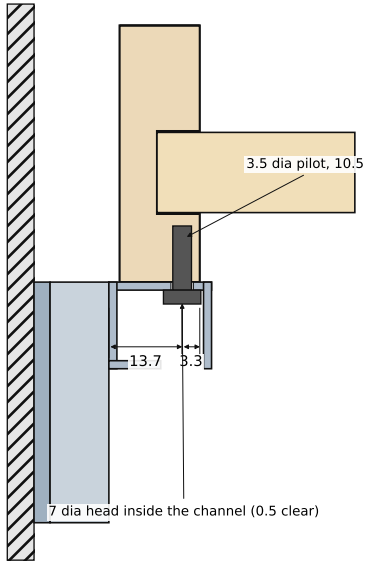
A: TYPICAL SECTION

through a side, at any point along the box



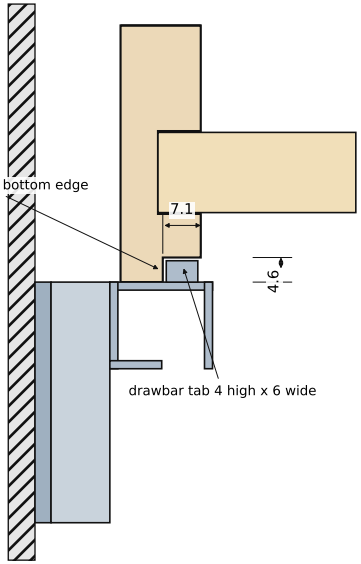
B: OPTION 2, SCREW FROM BELOW

drawbar hole 4.2 dia; screw No.6 x 12



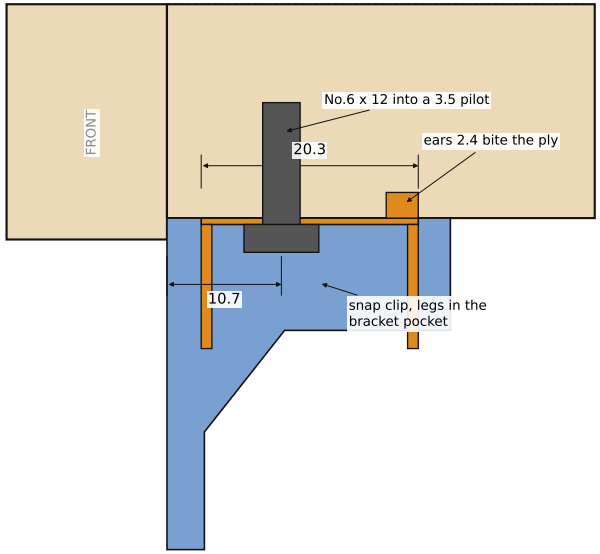
C: OPTION 1, REAR TAB IN THE NOTCH

section through the tab (481.9 from the front end)



D: OPTION 1, FRONT CLIP

longitudinal section, scale 2:1



The drawbar section is taken from the supplied STEP file; the bracket outline in D is approximate. The fixed runner block is shown as 14 x 45 (steel stack, true section not modelled).

Drawbar outer face is 14 from the cabinet wall (11 to the fixing plates + 3 plate). The 8 mm underside opening faces away from the wall, and the rear tab sits 3.9 toward it from the drawbar centre.

The side sits on the 15.2 flat of the drawbar, its outer face 16 from the wall. Left and right assemblies are mirror images. Option 1 notch is on the inner face; option 2 pilots are 3.3 from the inner face.

Choose one option per build (DB_FIX in the model). Option 1 needs no cutting of the steel; option 2 needs the tabs ground off and 2 holes per drawbar (36.5 behind the bracket, 38 from the rear end).

KITCHEN BENCH EXTENSION - EXTENSION UNIT

DRAWER BOX ON THE DRAWBARS: SECTIONS AND FIXING OPTIONS

Scale 1:1 (sections), 15 mm ply sides, drawbar 19.2 wide from the STEP file

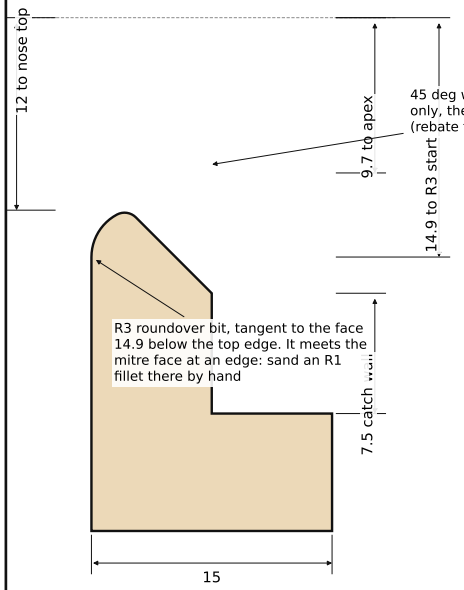
15 mm ply unless noted. Do not scale: dimensions govern.

SHEET **8**

DATE 2026-11-07

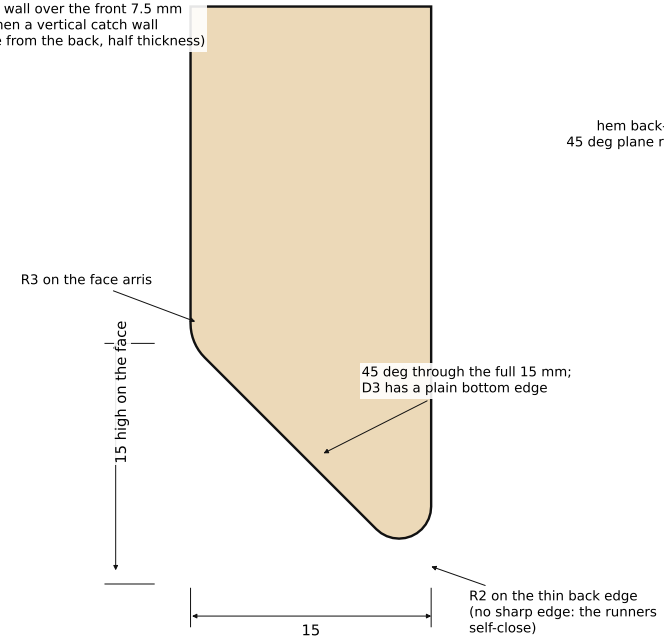
A: TOP EDGE, 45 DEG MITRE WITH CATCH WALL (D1, D2, D3)

section through a front, 3:1



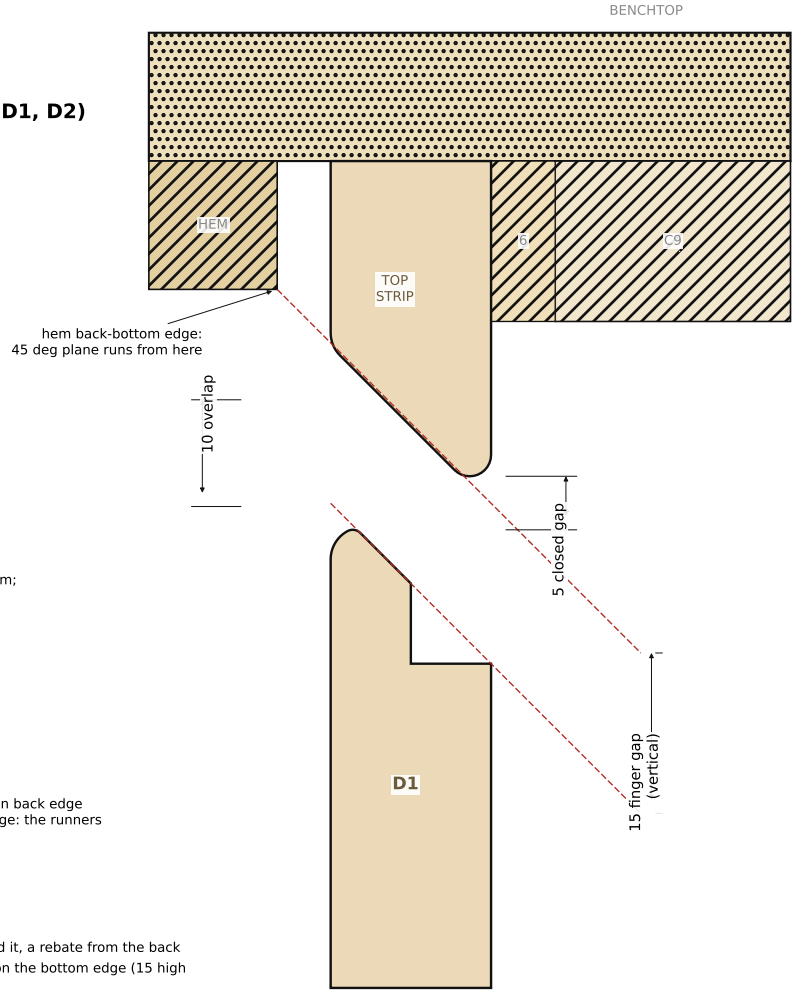
A2: BOTTOM EDGE, 45 DEG MITRE (D1, D2)

section through a front, 3:1



C: THE STACK AT THE TOP (D1 UNDER THE TOP STRIP)

section through the front edge, 2:1; D2 sits under D1 in the same way



Pulls: every front has a 45 degree mitre on the front 7.5 mm of its top edge (R3 nose, R1 sanded) and a vertical 7.5 mm catch wall behind it, a rebate from the back at half thickness, down to the shelf 24.7 below the top edge. The fingertip hooks the catch wall. D1 and D2 also have a 45 degree mitre on the bottom edge (15 high on the face, R3 and R2 rounded). D3 has a plain bottom edge.

Fingertips go into the gap between the bottom mitre of a front (or of the top strip) and the top mitre of the front below, then hook the catch wall. The faces are parallel, 15 mm apart vertically (about 10.5 square to the faces). Measured with calipers: 15 gets a finger past the first knuckle, 10 only the tip.

Each nose stays 5 mm below the lowest point of the front above (after rounding the thin edge) and fronts overlap 10 mm, so a drawer cannot catch the one below as it opens. The fronts fix to the boxes through 6 mm oversize holes (about 1 mm of play each way), set on 2 mm spacers.

D1's top edge is placed so that its mitre plane is 15 mm (vertical) below a parallel 45 degree plane through the back-bottom edge of the benchtop hem. The top strip (802 x 32.3, glued to a 6 mm packer on C9) has the same bottom mitre as D1 and D2.

Rout the rebate from the back first (straight bit, 7.5 wide, to the shelf), then cut the 45 degree lip with the chamfer cutter; an optional 14 degree dovetail pass into the catch wall undercuts it. Bottom mitres on a tilted saw or with the chamfer bit and a fence. Round the R3 and R2 with roundover bits, then sand the R1 edge by hand. Test the profile on an offcut first; the nose is a 45 degree wedge: handle with care.

KITCHEN BENCH EXTENSION - EXTENSION UNIT

MITRE PULL PROFILES AND GAPS

45 deg mitre pulls: top edge of D1 to D3, bottom edge of D1, D2 and the top strip. Sections at 3:1 and 2:1

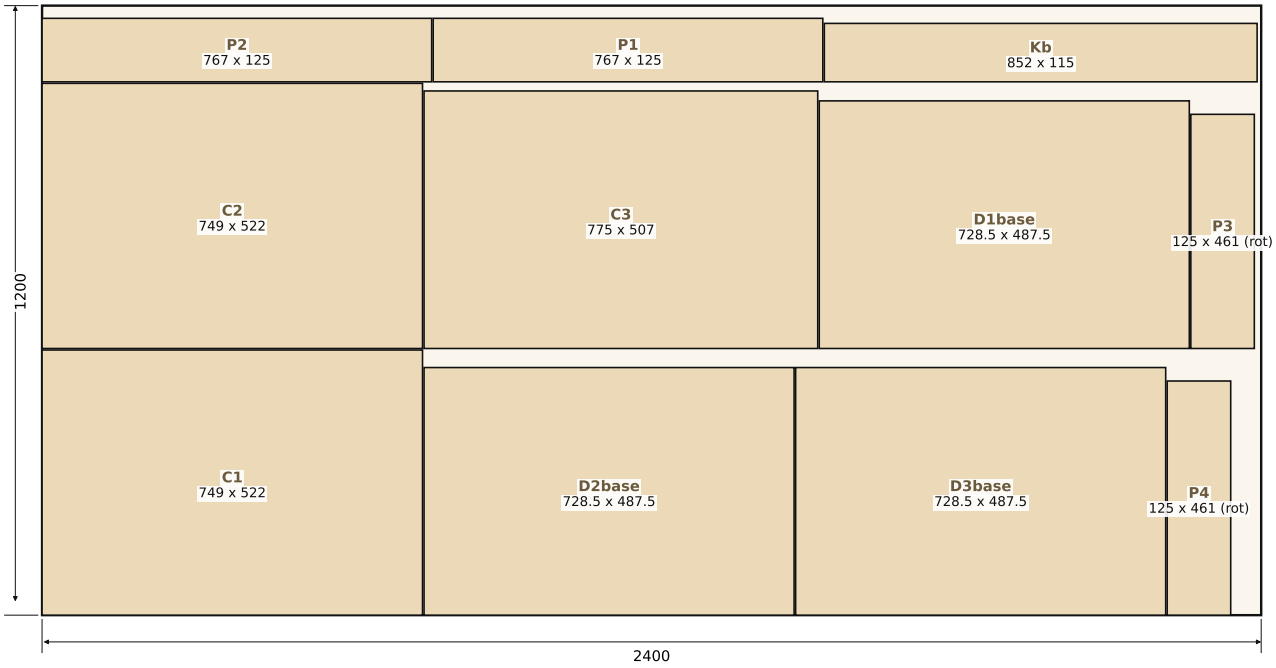
15 mm ply unless noted. Do not scale: dimensions govern.

SHEET 9

DATE 2026-11-07

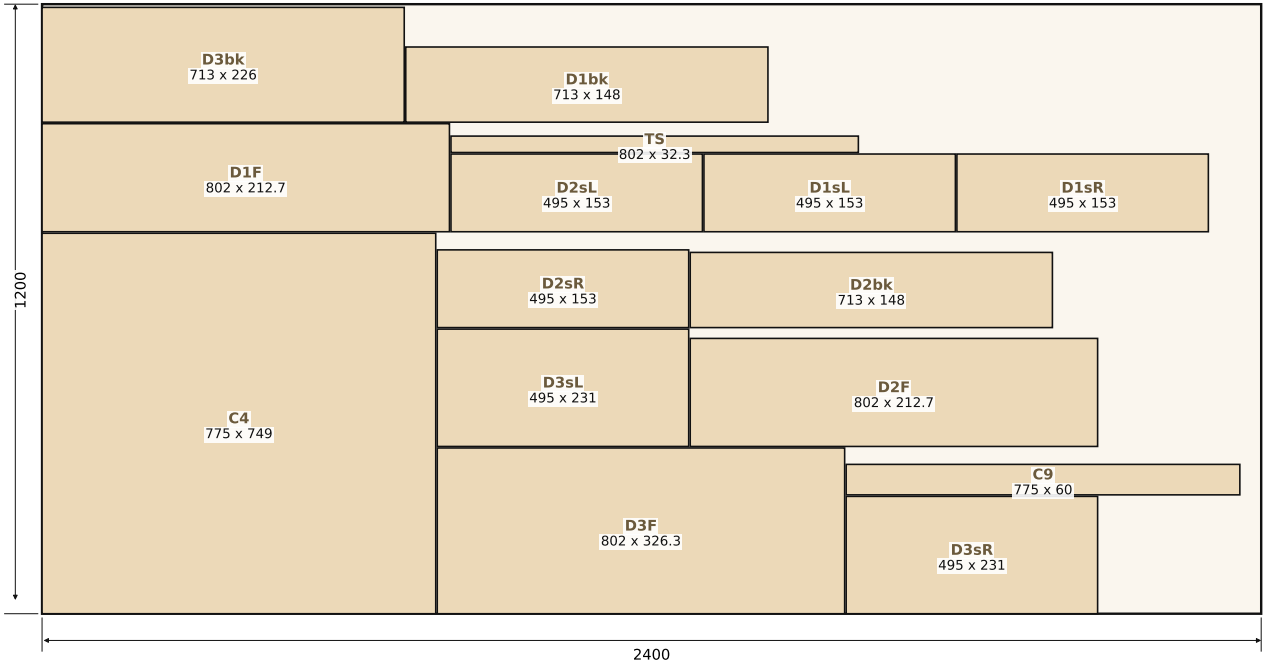
SHEET A

fill 92%



SHEET B

fill 75%



Grain: fronts (D1F to D3F) and the kickboard cover (Kb) are kept with the long edge along the sheet length.

Other parts may be rotated to fit, so the nesting is tight (about 83% of two sheets). Check it before cutting; if the grain must run the long way on every part, a third sheet may be needed.

Bin drawer, bay panels and sliding door (not part of the extension unit) are cut from a third sheet.

Sheet 11 is the alternative layout for a panel saw that only cuts on 100 mm multiples.

Label: Cx carcass/plinth; Dnx drawer n: F front, sL/sR sides, bk back, base. 30 mm of clearance around each part is not shown.

CUT LIST (15 mm ply, as cut; every piece is also placed on the diagrams)

Qty	Length	Width	Parts (sheet)
1	775	749	C4 (B)
1	775	507	C3 (A)
2	749	522	C1 (A), C2 (A)
3	728.5	487.5	D1base (A), D2base (A), D3base (A)
1	802	326.3	D3F (B)
2	802	212.7	D1F (B), D2F (B)
1	713	226	D3bk (B)
2	495	231	D3sL (B), D3sR (B)
2	713	148	D1bk (B), D2bk (B)
1	852	115	Kb (A)
2	767	125	P1 (A), P2 (A)
4	495	153	D1sL (B), D1sR (B), D2sL (B), D2sR (B)
2	461	125	P3 (A), P4 (A)
1	775	60	C9 (B)
1	802	32.3	TS (B)

26 pieces from 2 sheets of 2400 x 1200 x 15. Length is the first number; the 3 mm kerf is not included in these sizes.

KITCHEN BENCH EXTENSION - EXTENSION UNIT

CUTTING DIAGRAMS: EXTENSION UNIT

2 sheets of 2400 x 1200 x 15 ply. Scale 1:10. 3 mm kerf allowed

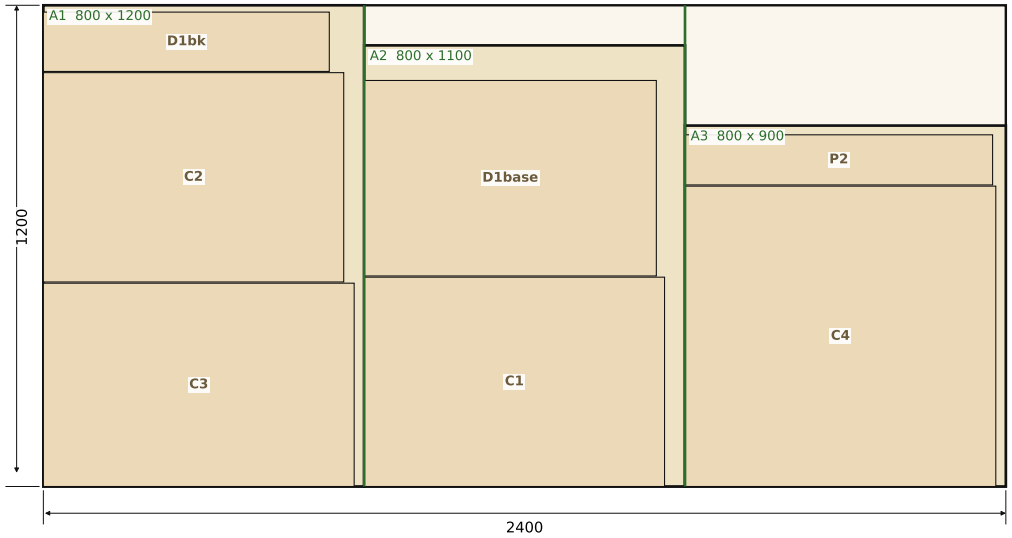
15 mm ply unless noted. Do not scale: dimensions govern.

SHEET **10**

DATE 2026-11-07

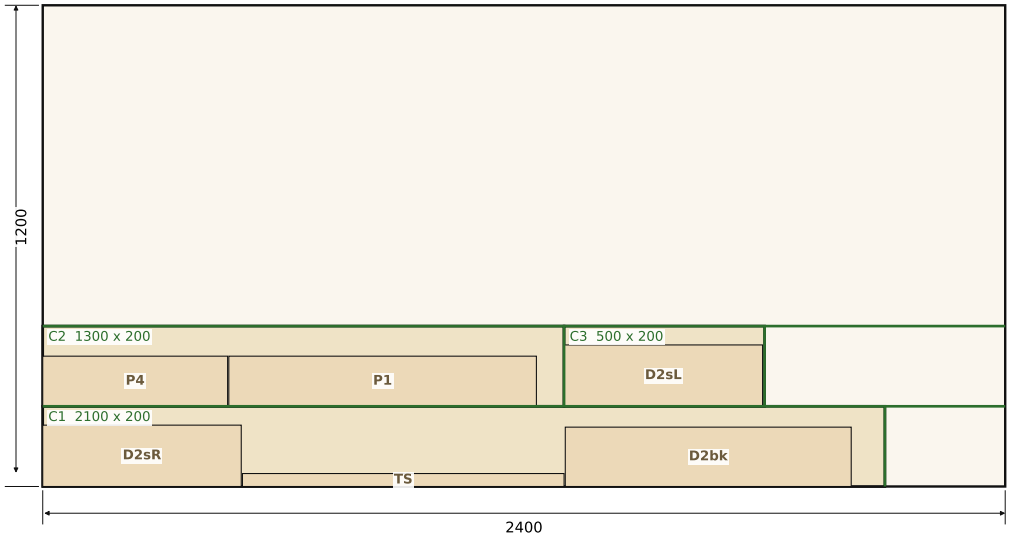
SHEET A

blanks fill 100% of the sheet



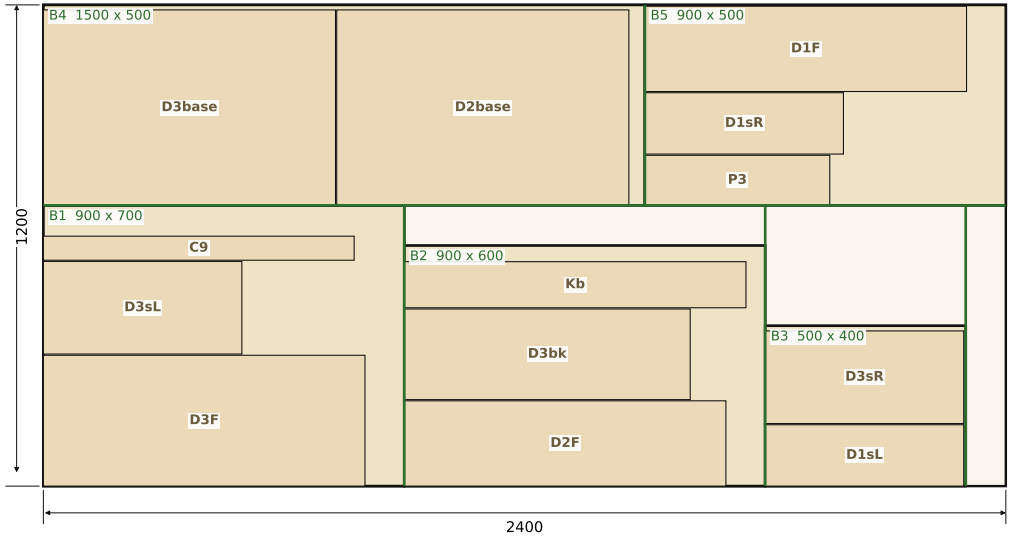
SHEET C

blanks fill 27% of the sheet



SHEET B

blanks fill 98% of the sheet



STORE CUTS (all on multiples of 100, measured from the sheet edge or the last cut face)

Sheet A: rip at none (single strip) from the long edge
Sheet A strip 1 (1200 wide): cross-cut at 800, 1600
Sheet B: rip at 700 from the long edge
Sheet B strip 1 (700 wide): cross-cut at 900, 1800, 2300
Sheet B strip 2 (500 wide): cross-cut at 1500
Sheet C: rip at 200, 400 from the long edge
Sheet C strip 1 (200 wide): cross-cut at 2100
Sheet C strip 2 (200 wide): cross-cut at 1300, 1800

BLANKS AND WHAT TO TRIM AT HOME (green = cut at the store; stacked parts are separated at home, 3 mm kerf allowed)

A1 (800 x 1200): C3 775 x 507 + C2 749 x 522 + D1bk 713 x 148
A2 (800 x 1100): C1 749 x 522 + D1base 728.5 x 487.5
A3 (800 x 900): C4 775 x 749 + P2 767 x 125
B1 (900 x 700): D3F 802 x 326.3 + D3sL 495 x 231 + C9 775 x 60
B2 (900 x 600): D2F 802 x 212.7 + D3bk 713 x 226 + Kb 852 x 115
B3 (500 x 400): D1sL 495 x 153 + D3sR 495 x 231
B4 (1500 x 500): D3base 728.5 x 487.5 + D2base 728.5 x 487.5
B5 (900 x 500): P3 461 x 125 + D1sR 495 x 153 + D1F 802 x 212.7
C1 (2100 x 200): D2sR 495 x 153 + TS 802 x 32.3 + D2bk 713 x 148
C2 (1300 x 200): P4 461 x 125 + P1 767 x 125
C3 (500 x 200): D2sL 495 x 153

Every part is cut oversize to a 100 mm blank and trimmed at home; blanks that hold two or more parts need a home cut between them. Grain: D1F to D3F and Kb stay with the long edge along the sheet. The saw kerf (about 3 mm) is taken from the waste side. Sheet C has room left for the bin drawer, bay and door parts.

KITCHEN BENCH EXTENSION - EXTENSION UNIT

CUTTING DIAGRAMS: STORE CUTS ON 100 MM MULTIPLES

3 sheets of 2400 x 1200 x 15 ply. Scale 1:13.
Alternative to sheet 10 if the store only cuts to 100 mm

15 mm ply unless noted. Do not scale: dimensions govern.

SHEET 11

DATE 2026-11-07